General Bite #4: Special Propagation -- Scatter, NVIS, Polarization

Ionospheric regions, critical frequency and critical angle, HF scatter into the skip zone, NVIS for short-range HF, and D-region daytime absorption.

Group G3C · 11 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

The Four Ionospheric Regions

The ionosphere has named layers stacked above Earth. From bottom to top: **D, E, F1, F2**.

Region	Behavior	Q
D	**Closest to Earth's surface** (G3C01). Daytime only. Absorbs HF signals (especially low HF) rather than refracting them.	G3C01
E	Lower-altitude refracting layer; short-hop daytime propagation, plus sporadic-E.	—
F1	Daytime-only intermediate layer.	—
F2	Highest refracting region. Persists day and night. Carries most long-distance HF.	—

Critical Frequency And Critical Angle

Critical frequency: The highest frequency which is refracted back to Earth at a given incidence angle (G3C02). Above the critical frequency, the wave punches through.

Critical angle: The highest takeoff angle that will return a radio wave to Earth under specific ionospheric conditions (G3C04). Above the critical angle, the wave punches through.

Why F2 Skip Goes Furthest

Why is skip propagation via the F2 region longer than via the other regions? Because it is the highest (G3C03). A higher-altitude refraction point geometrically gives a longer one-hop distance.

D-Region Daytime Absorption

Why is long-distance communication on 40, 60, 80, and 160 m more difficult during the day? The D region absorbs signals at these frequencies during daylight hours (G3C05).

Also: **which ionospheric region is the most absorbent of signals below 10 MHz during daylight? The D region** (G3C11). Same fact, two questions.

Practical implication: the low bands open up after dark when the D region dissolves.

HF Scatter

HF scatter is propagation that puts signals into the skip zone (the dead zone between ground wave and the first skip return) by scattering some energy off ionospheric irregularities.

Question	Answer	Q
Characteristic of HF scatter signals	**Signals have a fluttering sound** (G3C06).	G3C06
Why scatter signals sound distorted	**Energy is scattered into the skip zone through several different paths** (G3C07). Different path lengths arrive at slightly different times → smearing.	G3C07
Why scatter signals in the skip zone are weak	**Only a small part of the signal energy is scattered into the skip zone** (G3C08).	G3C08
Propagation that lets signals be heard in the transmitting station's skip zone	**Scatter** (G3C09).	G3C09

NVIS: Near-Vertical Incidence Skywave

NVIS = Short-distance MF or HF propagation at high elevation angles (G3C10).

Fire a low-band signal nearly straight up; the F layer (F1 or F2) reflects it back down to cover a 0-300 mile radius circle with no skip zone. Used for tactical comms, EMCOMM, and any time you need reliable short-range HF coverage without VHF infrastructure.

Mental Cheat Sheet

- Region nearest Earth = **D region**.
- D region absorbs LF/HF during the day.
- 40/60/80/160 m daytime difficulty = **D region absorption**.
- F2 is **highest**, gives longest skip.
- Critical frequency = highest f that refracts.
- Critical angle = highest takeoff angle that returns.
- HF scatter = **fluttering** sound.

Common Traps

- Closest region is **D**, not E or F.
- F2 skip is longest because of **altitude**, not density or Doppler.
- The "highest frequency refracted" is the **critical** frequency (specific to incidence angle).
- Critical angle is the **highest** takeoff angle (not the lowest).

- Scatter weak because **only a small fraction** of energy scatters into the skip zone.
- Scatter distorted because of **multiple path lengths**.
- Scatter is what fills the skip zone.
- NVIS = short-range MF/HF at **high elevation angles**.

- Daytime low-band trouble = **D absorption**, not F-region instability.
- HF scatter fills the **skip zone**, not the ground-wave zone.
- NVIS is **short-distance** propagation, not long-distance.
- NVIS uses **high** elevation angles, not low ones.

Takeaway: The ionosphere is a four-layer cake: D (daytime absorber), E (short hops), F1 (daytime helper), F2 (the long-haul layer). Critical frequency tells you the highest frequency that refracts; critical angle tells you the highest angle that returns to Earth. Scatter fills the skip zone with weak fluttering signals via multiple paths. NVIS fires straight up to cover 0-300 miles with no dead zone.

Bite 4 Question Bank (G3C)

G3C01. Which ionospheric region is closest to the surface of Earth?
A. The D region.
B. The E region.
C. The F1 region.
D. The F2 region.

G3C02. What is meant by the term “critical frequency” at a given incidence angle?
A. The highest frequency which is refracted back to Earth.
B. The lowest frequency which is refracted back to Earth.
C. The frequency at which the signal-to-noise ratio approaches unity.
D. The frequency at which the signal-to-noise ratio is 6 dB.

G3C03. Why is skip propagation via the F2 region longer than that via the other ionospheric regions?
A. Because it is the densest.
B. Because of the Doppler effect.
C. Because it is the highest.
D. Because of temperature inversions.

G3C04. What does the term “critical angle” mean, as applied to radio wave propagation?
A. The long path azimuth of a distant station.
B. The short path azimuth of a distant station.
C. The lowest takeoff angle that will return a radio wave to Earth under specific ionospheric conditions.
D. The highest takeoff angle that will return a radio wave to Earth under specific ionospheric conditions.

G3C05. Why is long-distance communication on the 40-, 60-, 80-, and 160-meter bands more difficult during the day?
A. The F region absorbs signals at these frequencies during daylight hours.
B. The F region is unstable during daylight hours.
C. The D region absorbs signals at these frequencies during daylight hours.
D. The E region is unstable during daylight hours.

G3C06. What is a characteristic of HF scatter?
A. Phone signals have high intelligibility.
B. Signals have a fluttering sound.
C. There are very large, sudden swings in signal strength.
D. Scatter propagation occurs only at night.

G3C07. What makes HF scatter signals often sound distorted?
A. The ionospheric region involved is unstable.
B. Ground waves are absorbing much of the signal.
C. The E region is not present.
D. Energy is scattered into the skip zone through several different paths.

G3C08. Why are HF scatter signals in the skip zone usually weak?
A. Only a small part of the signal energy is scattered into the skip zone.
B. Signals are scattered from the magnetosphere, which is not a good reflector.
C. Propagation is via ground waves, which absorb most of the signal energy.
D. Propagation is via ducts in the F region, which absorb most of the energy.

G3C09. What type of propagation allows signals to be heard in the transmitting station’s skip zone?
A. Faraday rotation.
B. Scatter.
C. Chordal hop.
D. Short-path.

G3C10. What is near vertical incidence skywave (NVIS) propagation?
A. Propagation near the MUF.
B. Short distance MF or HF propagation at high elevation angles.
C. Long path HF propagation at sunrise and sunset.
D. Double hop propagation near the LUF.

G3C11. Which ionospheric region is the most absorbent of signals below 10 MHz during daylight hours?
A. The F2 region.
B. The F1 region.
C. The E region.
D. The D region.

Answer Key

G3C01: A	G3C02: A	G3C03: C	G3C04: D
G3C05: C	G3C06: B	G3C07: D	G3C08: A
G3C09: B	G3C10: B	G3C11: D	